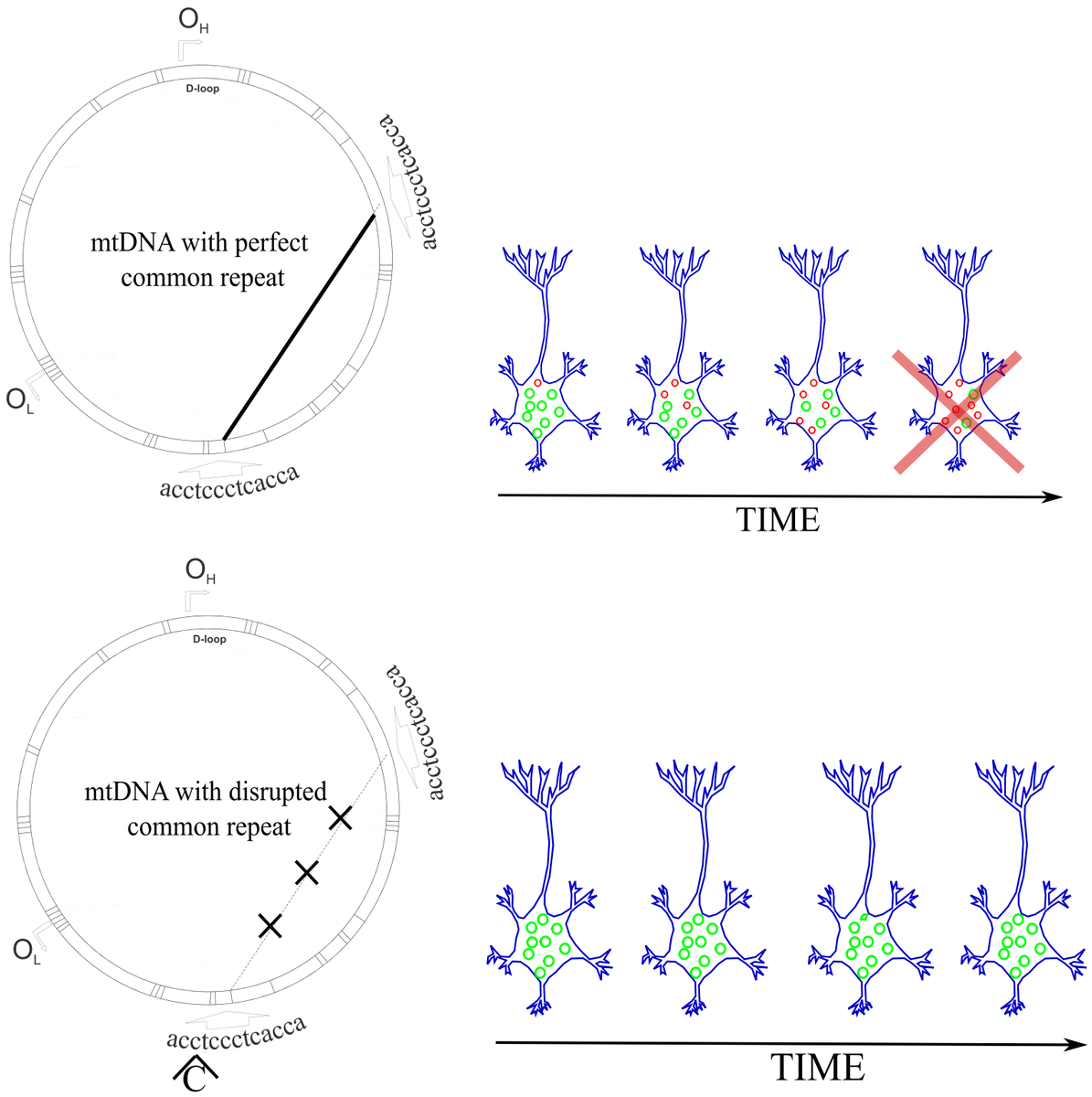
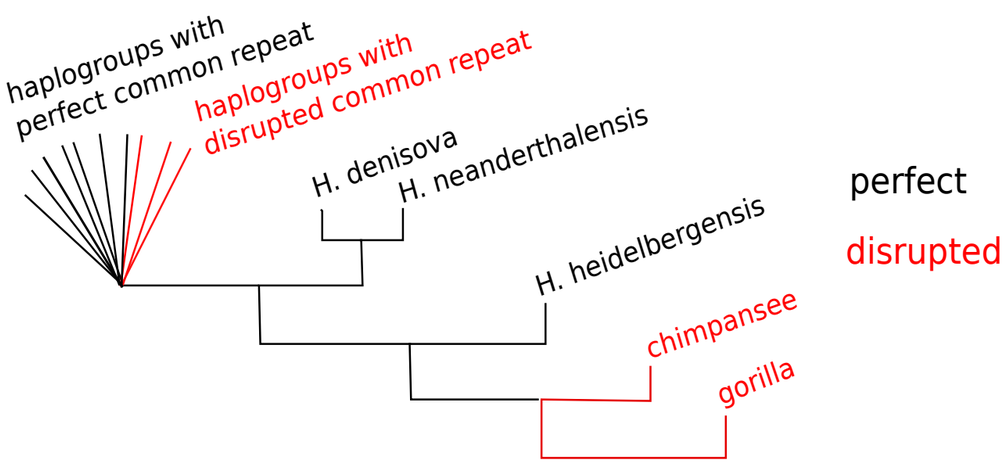
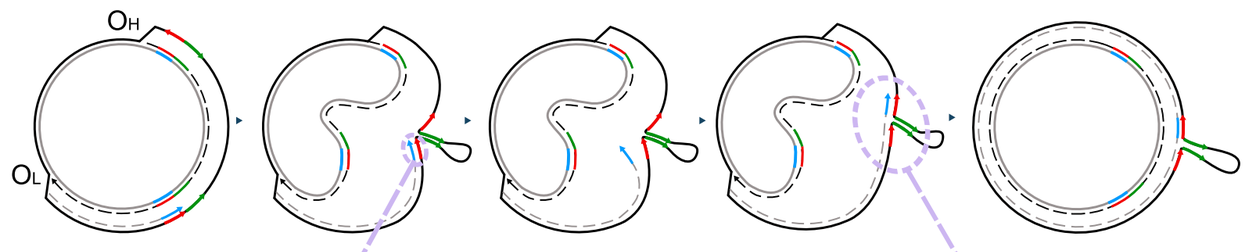
A.


O_H
D-loop
acctcccacca
mtDNA with perfect common repeat
O_L
acctccctcacca
TIME
O_H
D-loop
acctcccacca
mtDNA with disrupted common repeat
O_L
acctccctcacca
C
TIME

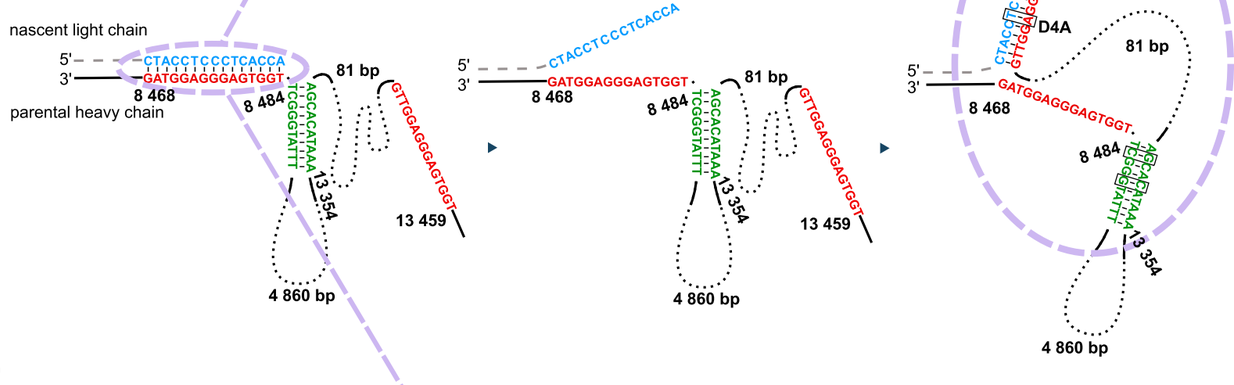
B.


haplogroups with perfect common repeat
haplogroups with disrupted common repeat
H. denisova
H. neanderthalensis
H. heidelbergensis
chimpansee
gorilla
perfect
disrupted

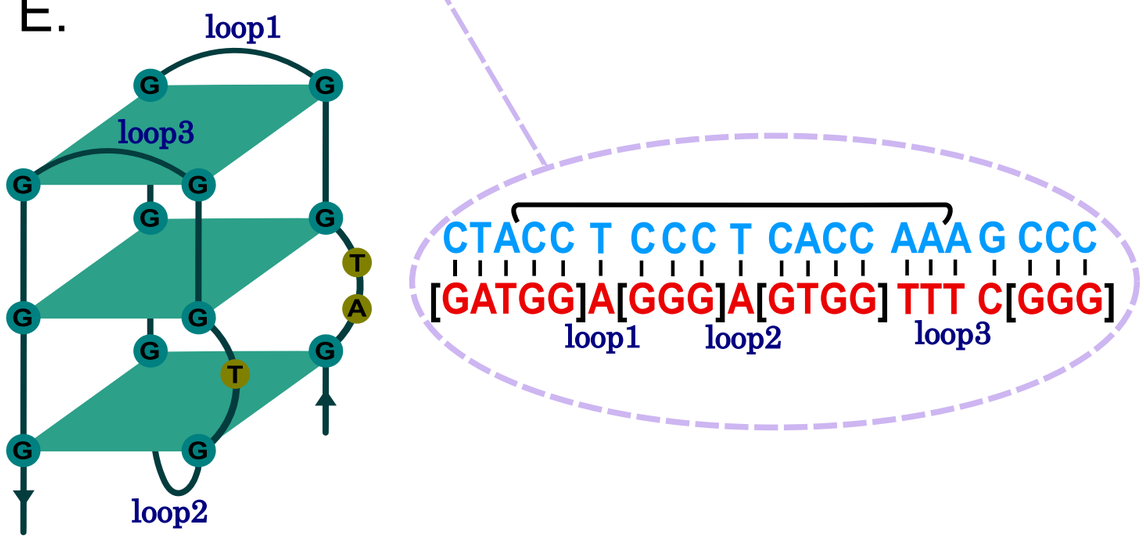
C.


O_H
O_L

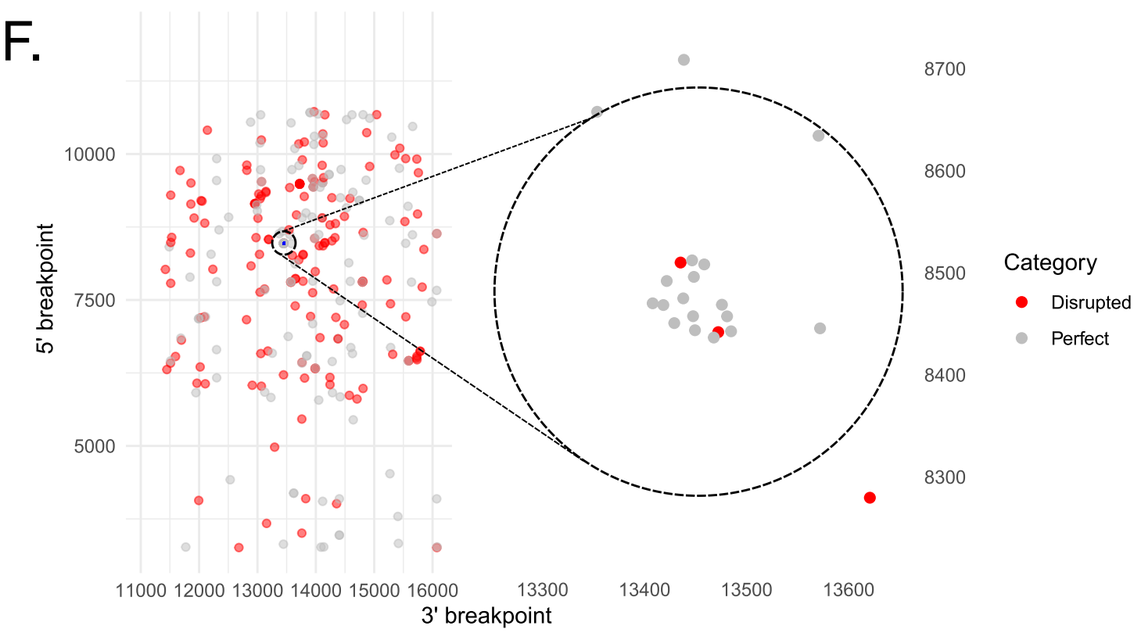
D.


nascent light chain
5'
3'
CTACCTCCCTCACCA
GATGGAGGGAGTGGT
8 468
parental heavy chain
8 484
81 bp
AGCACATAAA
TCGGGTATTT
13 354
GTTGGAGGGAGTGGT
13 459
4 860 bp
CTACCTCCCTCACCA
GATGGAGGGAGTGGT
D5A
D4A

E.


loop1
loop3
loop2
G
T
A
CTACC T CCC T CACC AAA G CCC
[GATGG]A[GGG]A[GTGG] TTT C[GGG]
loop1
loop2
loop3

F.


5' breakpoint
10000
7500
5000
11000 12000 13000 14000 15000 16000
3' breakpoint
13300 13400 13500 13600
8700
8600
8500
8400
8300
Category
Disrupted
Perfect